

AP CSP | Lesson 10: Boolean Logic

Homework | Multiple-choice and written practice

Name: _____ Class: _____ Date: _____

Time: 40 minutes | Total: 30 teacher marks (not an AP score) | Show working.

Original AP-style MCQs and classroom written practice; these are not official AP questions.

Q5: select exactly TWO answers. OR is inclusive. All named logical operands are Booleans; age and score values are numbers. Parentheses specify grouping.

1. What are $14 / 5$ and $14 \text{ MOD } 5$, respectively? [1 mark]

- A. 2 and 4 B. 2.8 and 2 C. 2.8 and 4 D. 4 and 2.8

Answer: _____

2. Which expression is true? [1 mark]

- A. $8 < 8$ B. $8 \neq 8$ C. $8 > 8$ D. $8 \leq 8$

Answer: _____

3. Trace: $\text{score} \leftarrow 7$; $\text{passed} \leftarrow (\text{score} \geq 5)$; $\text{score} \leftarrow 2$. What does passed hold? [1 mark]

- A. true B. false
C. 2 D. The expression $\text{score} \geq 5$, which updates automatically

Answer: _____

4. booked and hasPass are both true. What is booked OR hasPass? [1 mark]

- A. false, because exactly one must be true B. true, because OR allows one or both true
C. 2, because two conditions are true D. A string containing "true"

Answer: _____

5. For Boolean A and B, which TWO statements are true? Select exactly TWO answers. [1 mark]

- A. true AND false evaluates to true. B. NOT false evaluates to true.
C. NOT (A OR B) always equals (NOT A) OR (NOT B). D. NOT (A AND B) always equals (NOT A) OR (NOT B).

Answer: _____

6. Which expression is true exactly when age is from 12 through 17, inclusive? [1 mark]

- A. $(\text{age} > 12) \text{ AND } (\text{age} < 17)$ B. $(\text{age} \geq 12) \text{ OR } (\text{age} \leq 17)$
C. $(\text{age} \geq 12) \text{ AND } (\text{age} \leq 17)$ D. $(\text{age} = 12) \text{ OR } (\text{age} = 17)$

Answer: _____

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Written responses | Questions 7-9

7.

[4 marks]

For score = 20, evaluate each expression.

- (a) score > 20 (b) score $\geq$ 20
(c) score $\neq$ 20 (d) NOT (score > 20)

8.

[4 marks]

For each row, give A AND B and A OR B, in that order.

- (a) A=true, B=true (b) A=true, B=false
(c) A=false, B=true (d) A=false, B=false

9.

[4 marks]

Write a Boolean expression for each condition.

- (a) x is at least 4 and at most 9.
(b) x is outside that inclusive range, without using NOT.
Then evaluate expression (a) for (c) x=4 and (d) x=10.

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Written responses | Questions 10-12

10. [4 marks]

(a) Rewrite NOT (A AND B) with NOT applied only to individual operands.

(b) Do the same for NOT (A OR B).

For A=false and B=true, evaluate (c) NOT (A AND B) and (d) (NOT A) AND (NOT B).

11. [4 marks]

Use $\text{inRange} \leftarrow (\text{age} \geq 12) \text{ AND } (\text{age} \leq 17)$, $\text{hasEntry} \leftarrow \text{booked OR hasPass}$, $\text{eligible} \leftarrow \text{inRange AND hasEntry}$. For age=14, booked=true, hasPass=true, give (a) inRange, (b) hasEntry, (c) eligible. (d) On a new run with age=18 and the same credentials, give eligible.

12. [4 marks]

Bug: $\text{inRange} \leftarrow (\text{age} \geq 12) \text{ OR } (\text{age} \leq 17)$. The intended range is 12-17 inclusive.

(a) Give the buggy result for age=5. (b) Correct the assignment.

For the corrected code give the expected result at (c) age=12 and (d) age=11.
